

Recent Trends in Secure Key Distribution and Key Management for Modern Network Security: A Literature Review

Aryan Jotshi, Tanmay Karad, Tanishq Borkar, Arnav Singhal

Abstract

Secure key distribution underlies almost every mechanism used to protect data in transit, and the field has moved quickly in recent years. This paper reviews the evolution of key distribution and key management, starting from the classical Diffie-Hellman and RSA-based foundations, through authentication-centered protocols such as Kerberos, Public Key Infrastructure (PKI), TLS 1.3, and IPsec/SSH, and on to several major developments between roughly 2023 and 2026: the NIST post-quantum cryptography standards and their hybrid deployment in TLS, the maturing state of quantum key distribution, blockchain-based approaches to decentralizing public key infrastructure, and the growing use of machine learning for adaptive key-lifecycle management. We also examine how these techniques apply, and where they still struggle, in IoT, cloud, 5G/6G, and vehicular settings. Across the literature surveyed, no single technique replaces the others. Production systems increasingly combine them, and formally proven security guarantees remain the clearest gap in the newer, less mature approaches.

Index Terms

Key distribution, key management, Diffie-Hellman, public key infrastructure, TLS 1.3, post-quantum cryptography, quantum key distribution, blockchain, zero trust, network security.

I. INTRODUCTION

Every cryptographic protection scheme, whether it encrypts an online transaction or authenticates a firmware update, ultimately rests on a shared secret that two or more parties must obtain safely. This is the key distribution problem, and it has been part of network security since Diffie and Hellman's 1976 proposal for exchanging a secret over a public channel [1]. In the decades since, the problem has not gone away so much as it has multiplied. Key distribution now has to work across enterprise domains, the open Internet, and resource-constrained IoT devices, and it must do so under the shadow of quantum computers that threaten the mathematics the field has relied on for fifty years.

This paper is divided into five sections, each covering a distinct part of the problem. Section II covers the classical foundations: manual and trusted-third-party key distribution, Diffie-Hellman and RSA-based exchange, and the formal security frameworks that later justified combining them with authentication. Section III moves to the authentication-centered protocols that dominate deployed systems today, namely Kerberos, Public Key Infrastructure (PKI), the TLS 1.3 handshake, and IPsec/SSH. Section IV reviews four developments that have matured between 2023 and 2026: standardized post-quantum cryptography and hybrid key exchange, quantum key distribution and the early quantum internet, blockchain-based decentralized key and certificate management, and AI-assisted key-lifecycle management. Section V examines how these ideas play out in IoT, cloud, edge, 5G/6G, and vehicular networks, together with the open challenges that remain.

Throughout, we try to separate what is reasonably well established in the literature from what is still an open question, and we note where a claimed benefit, for instance an "AI-assisted" or "post-quantum" label, has not yet been backed by a formal security proof.

II. FUNDAMENTALS OF KEY DISTRIBUTION

A cryptographic key is the piece of secret information that parameterizes an encryption or authentication function. Because the strength of a cipher reduces to the secrecy of its key, distributing keys safely has been treated as a first-class problem since the earliest work on network cryptography. A key normally passes through a lifecycle of generation, distribution, activation, use, rotation, and eventual revocation or destruction, and standardized key-management guidance such as NIST SP 800-57 is largely concerned with securing each of these stages [9]. Keys are usually classified as symmetric (a single shared secret, as in AES) or asymmetric (a mathematically related public/private pair), and as long-term identity keys versus short-lived session keys.

The earliest approach, manual key distribution, delivered keys out of band, by courier or keyed hardware. It offers strong assurance but does not scale: the number of shared symmetric keys needed grows quadratically with the number of parties ($n(n-1)/2$ for n parties), and any compromise requires redistribution through the same slow channel. Symmetric distribution via a trusted third party (TTP), later formalized in Needham-Schroeder and Kerberos (Section III), reduces this to one long-term key per party plus a central key distribution center (KDC), at the cost of a single point of trust and failure.

TABLE I
CLASSICAL KEY DISTRIBUTION TECHNIQUES

Technique	Authentication	Computational Cost	Forward Secrecy
Manual distribution	Out-of-band trust	Very low	N/A
TTP / KDC	Implicit (master key)	Low	Policy-dependent
Unauthenticated DH	None; MITM-exposed	Moderate	Yes (fresh exponents)
Authenticated DH / STS	Signatures / MACs	Moderate to high	Yes, if keys erased
RSA key transport	Certificates / PKI	High	No, unless paired with ephemeral DH

The Diffie-Hellman (DH) exchange was the decisive classical breakthrough. It lets two parties with no prior shared secret agree on a key over a public channel, with security resting on the hardness of the discrete-logarithm problem [1]. Unauthenticated DH, however, does not bind the exchanged values to identities, so it is fully exposed to an active man-in-the-middle adversary. This motivated authenticated key exchange, most notably the Station-to-Station (STS) protocol, which layers digital signatures over DH so that each party proves possession of a long-term signing key while negotiating the session key. Formal frameworks such as the Bellare-Canetti-Krawczyk authenticator compilation and the related Canetti-Krawczyk session-key (SK-)security definition later provided theoretical justification for composing authenticated key exchange with symmetric encryption to build secure channels [2], [3].

RSA-based key transport, historically used in TLS prior to TLS 1.3, took a different route: instead of jointly deriving a key, one party generates a session key and encrypts it directly under the other party's RSA public key, relying on a certificate authority to bind the public key to an identity. Formal methods have also played a foundational role. Meadows's analysis of the Internet Key Exchange (IKE) protocol with the NRL Protocol Analyzer found specification ambiguities, in particular unclear binding of keys to identities versus IP addresses, that made practical attacks possible even in a well-vetted, standardized protocol [4].

None of this is settled history. NIST's key-management guidance continues to be revised: 1024-bit DH and RSA moduli are no longer approved, 2048-bit is treated as the practical minimum at 112-bit security strength, and 3072-bit or larger is recommended for longer-term protection [9]. At Internet scale, hybrid classical/post-quantum groups (discussed in Section IV) are now negotiated alongside plain elliptic-curve DH in a substantial share of TLS connections, which suggests classical DH is being extended rather than retired in current practice [14].

The trade-off in Table I is a recurring theme in this review: techniques that minimize trust assumptions sacrifice authentication, while techniques that add authentication add computational or infrastructural cost. The Bellare-Canetti-Krawczyk and Canetti-Krawczyk results matter because they show this trade-off is not ad hoc: a protocol proven secure in an idealized model can be compiled into an authenticated one without re-deriving security from scratch [2], [3], which is the theoretical basis for the authentication-layered protocols discussed next.

III. AUTHENTICATION AND SECURE KEY DISTRIBUTION

Where Section II showed that unauthenticated key agreement is not enough on its own, this section reviews the protocols that add identity verification to key distribution in networks deployed today.

A. Kerberos

Kerberos, developed at MIT under Project Athena, is a symmetric-key, trusted-third-party protocol that still dominates authentication inside enterprise networks. A client authenticates once with an Authentication Server (AS) and receives a Ticket Granting Ticket (TGT); it then presents the TGT to a Ticket Granting Server (TGS) to obtain service-specific tickets, so passwords are never sent over the network and single sign-on follows naturally [5]. Its main weaknesses, dependence on synchronized clocks and a KDC that represents a logical single point of trust, are largely unchanged since the original 1994 description, but Kerberos is being extended rather than replaced. Modern enterprise identity systems continue to integrate Kerberos with cloud identity services such as Azure AD and Entra ID, allowing ticket issuance to interoperate with cloud identity providers in hybrid, zero-trust environments.

B. Public Key Infrastructure

PKI provides authenticated public-key distribution by binding a public key to a claimed identity through X.509 certificates issued by a Certificate Authority (CA) [6]. PKI underlies HTTPS, email security, VPN authentication, and code signing, but a centralized CA is itself a single point of failure, and certificate mis-issuance has repeatedly been a real incident rather than only a theoretical risk. Certificate Transparency addresses this by requiring certificates to be logged in public, append-only logs that domain owners can monitor, which helps detect mis-issuance after the fact rather than preventing it outright [7]. Certificate

TABLE II
AUTHENTICATION-BASED KEY DISTRIBUTION PROTOCOLS

Protocol	Authentication	Key Exchange	Primary Use
Kerberos	Tickets (AS/TGS)	Symmetric session key	Enterprise networks
PKI	X.509 certificates	Public-key cryptography	Internet PKI
TLS 1.3	Certificate + signature	Ephemeral DH	HTTPS
SSH	Host keys	DH / ECDH	Remote administration
IPsec/IKE	Certificates or PSK	DH	VPN

lifetimes are also being compressed for related reasons: the CA/Browser Forum has approved a phased schedule that reduces the maximum public TLS certificate lifetime from 398 days to 47 days by 2029, which makes automated issuance protocols such as ACME effectively mandatory rather than optional.

C. TLS 1.3, SSH, and IPsec

TLS 1.3 is the protocol most people interact with, knowingly or not, when they browse the web. Compared with TLS 1.2, it removes RSA key transport and static DH, mandates ephemeral Diffie-Hellman based key exchange, which provides forward secrecy, authenticates the handshake with a certificate and signature, and reduces the handshake to one round trip in the common case [8]. TLS 1.3 is also the vehicle for the hybrid post-quantum key exchange discussed in Section IV, since TLS 1.2 cannot host the new named groups.

SSH and IPsec extend the same idea to remote administration and network-layer VPNs, respectively. SSH authenticates the server via host keys and negotiates session keys with Diffie-Hellman or elliptic-curve DH; IPsec's IKE negotiates security associations and keys between gateways. As Section II noted, formal analysis of IKE found real specification gaps even though the protocol had been standardized and reviewed for years [4].

D. From Perimeter Authentication to Zero Trust

The clearest shift in this part of the literature is away from a one-time, perimeter-based authentication event and toward continuous verification. NIST SP 800-207 formalizes Zero Trust Architecture (ZTA) around the principle that no user or device is trusted by default, and that trust decisions are continuously evaluated using identity, device posture, and behavioral signals, rather than relying on a single authentication event at session start [10]. A 2024 survey maps ZTA specifically onto 6G networks, arguing that the device density and heterogeneity of 6G make static, perimeter-based trust unworkable [11]. Multi-factor authentication, combining a password with a token, biometric, or one-time code, remains the most widely deployed way to reduce the impact of credential theft, and current TLS and enterprise-authentication planning increasingly treats post-quantum readiness as part of the same migration rather than a separate later project [12].

IV. RECENT TRENDS IN KEY DISTRIBUTION

The classical and authenticated protocols reviewed above all rest on the hardness of factoring or the discrete-logarithm problem. Shor's algorithm solves both in polynomial time on a sufficiently large, fault-tolerant quantum computer, and because an adversary can record encrypted traffic today and decrypt it once such a machine exists (the "harvest now, decrypt later," or HNDL, threat), the 2023 to 2026 period has seen four developments move from proposal toward deployment: standardized post-quantum cryptography, maturing quantum key distribution, blockchain-based decentralization of PKI, and AI-assisted key management. None of these has replaced the others; recent systems increasingly combine them.

A. Post-Quantum and Hybrid Key Exchange

In August 2024, NIST finalized its first three post-quantum standards after an eight-year public evaluation [12]. ML-KEM (FIPS 203, derived from CRYSTALS-Kyber) is a module-lattice-based key-encapsulation mechanism with three parameter sets whose public keys and ciphertexts range from roughly 800 to 1568 bytes [13]. ML-DSA (FIPS 204, from CRYSTALS-Dilithium) is a lattice-based signature scheme with signatures of 2420 to 4627 bytes. SLH-DSA (FIPS 205, from SPHINCS+) is a conservative, stateless hash-based signature scheme whose security depends only on hash-function properties, but it is expensive: signatures range from 7856 to 49856 bytes, and signing is orders of magnitude slower than the lattice-based alternatives.

Because confidence in any single new hardness assumption is still being built, production systems deploy PQC in hybrid mode, pairing a classical Diffie-Hellman exchange with an ML-KEM encapsulation so that the shared secret stays safe if either component is later broken. The IETF has standardized this pattern for TLS 1.3, including named hybrid groups such as X25519MLKEM768 [14]. Google Chrome enabled a hybrid Kyber-based exchange by default from version 124 in April

2024, and Firefox followed with the standardized group. Several major Internet services have begun deploying hybrid PQC in TLS, though the share of overall HTTPS traffic negotiated this way is still comparatively small relative to classical ECDHE. Messaging platforms such as Signal (PQXDH) and Apple iMessage (PQ3) have likewise integrated Kyber-based hybrid key agreement into their end-to-end encryption. The measurable cost of this migration is larger handshakes: hybrid key shares and post-quantum certificates are substantially bigger than their classical ECDHE counterparts, with hash-based signatures adding the most latency.

B. Quantum Key Distribution

Unlike PQC, quantum key distribution (QKD) derives its security from physics rather than computational hardness. BB84 (1984) encodes bits in non-orthogonal photon polarization states across two conjugate bases; E91 (1991) instead distributes entangled photon pairs and certifies security through violation of a Bell/CHSH inequality. Recent work has focused on combining and hardening these protocols rather than replacing them. A 2025 framework integrates BB84, B92, E91, and GHZ-state protocols under a controller that selects the best-performing protocol in real time, reporting a CHSH value of $S = 2.5$ confirming entanglement fidelity; the authors describe it as the first fully integrated, AI-assisted hybrid QKD system [15].

Field deployment has also advanced. A 2025 urban-fiber trial combining one-decoy-state BB84 with qubit-based frame synchronization ran continuously for 12 hours, sustaining a secure key rate of 26.6 kbit/s at 18 dB channel loss and 115 bit/s even at 40 dB loss, without dedicated synchronization hardware [16]. At larger scales, China’s Micius satellite has demonstrated entanglement-based QKD over 1120 km, and a trusted-relay backbone linking Beijing and Shanghai spans roughly 2000 km of fiber [15].

A persistent open problem is authenticating the classical channel that accompanies every QKD link, traditionally done with pre-shared symmetric keys that scale quadratically with the number of network users. Recent work replaces this with post-quantum public-key authentication of the classical channel; an experimental demonstration authenticated a field QKD deployment entirely with post-quantum signatures [17]. The position that recurs in both the literature and industry commentary is that QKD and PQC are complementary rather than competing: QKD offers information-theoretic security for the key itself but needs specialized hardware and, over long distances, trusted relays or repeaters, while PQC is software-deployable at Internet scale but rests on hardness assumptions that, while extensively studied, are not proven.

C. Blockchain-Based Key and Certificate Management

The centralized CA model discussed in Section III has well-known weaknesses: a single point of failure, opaque revocation, and mis-issuance risk. Blockchain-based decentralized PKI (DPKI) records certificate issuance and revocation on a distributed ledger and uses smart contracts to automate the lifecycle without a single trusted CA. Several research proposals investigate this direction. These include PKChain, a compromise-tolerant, verifiable public-key management system built on blockchain and smart contracts that supports real-time key revocation and recovery [18], alongside a growing body of Ethereum- and Hyperledger-Fabric-based DPKI designs targeted at IoT identity and certificate management. Most of these systems remain research prototypes rather than production deployments; they generally manage certificate metadata, namely registration, revocation, and transparency, rather than performing the underlying key-exchange computation itself, and consensus latency together with on-chain storage cost remain the recurring bottleneck across this literature.

D. AI-Assisted Key Management

The final development is the use of machine learning to make key management adaptive rather than static. Recent research has explored AI-assisted approaches to key-lifecycle management, moving from fixed-schedule key rotation toward risk-driven rotation, where anomalous access patterns or elevated threat intelligence trigger rotation of specific keys instead of everything on a calendar. Applied examples include industrial-control-system frameworks that pair machine-learning anomaly detection with cryptographic key management, and reviews of AI-assisted cryptography that compare dynamic key generation and AI-enhanced cryptanalysis against classical and post-quantum baselines. This remains largely a research direction rather than an established practice: most proposed schemes still lack formal, reduction-based security proofs, and reported production deployment is limited compared with the PQC and QKD developments discussed above.

This limitation is worth stating plainly. Machine-learning techniques have also been explored by attackers targeting certificate-validation and key-distribution logic, and neither the blockchain- nor the AI-assisted schemes surveyed here yet offer security guarantees comparable to the modular authenticator framework and SK-security model introduced in Section II [2], [3]. As hybrid PQC, PQC-authenticated QKD, and blockchain- or AI-managed key lifecycles are combined in practice, Meadows’s original point about IKE, that suites of interacting subprotocols create vulnerabilities individual protocols do not exhibit alone [4], is arguably more relevant now than when it was written.

Taken together, the period covered here has diversified key distribution along four largely complementary axes rather than producing one winner. The clearest gap that recurs across this section is that blockchain- and AI-assisted key-management schemes, while operationally promising, still lack the formal, reduction-based security proofs that underpin classical and post-quantum key exchange.

TABLE III
COMPARISON OF EMERGING KEY-DISTRIBUTION APPROACHES

Approach	Security Basis	Quantum-Safe	Main Limitation
Hybrid PQC (ECDH + ML-KEM)	Lattice (MLWE) + discrete log	Yes (dual)	Larger handshakes, more computation
QKD (BB84 / E91)	Quantum mechanics	Yes (information-theoretic)	Specialized hardware; classical channel still needs authentication
Blockchain DPKI	Consensus + underlying crypto	Depends on primitive	Consensus latency; manages metadata only
AI-assisted key mgmt.	Heuristic / ML models	Not inherent	No formal security proofs yet

TABLE IV
KEY DISTRIBUTION TRADE-OFFS ACROSS DEPLOYMENT MODELS

Method	Security Focus	Cost	Quantum-Safe
Classical PKI	Centralized trust	Low	No
Hybrid PQC	HNDL resistance	High	Yes
Blockchain DPKI	Distributed consensus	High	Depends
Zero Trust	Continuous verification	Moderate	Depends

V. APPLICATIONS, CHALLENGES, AND FUTURE DIRECTIONS

The protocols and trends reviewed above were largely designed for relatively static topologies with a centralized trust anchor. IoT, cloud and edge computing, 5G/6G, and vehicular networks stretch that assumption in different ways: massive device counts, tight power and memory budgets, high mobility, and, for vehicles, service lifespans of fifteen to twenty years during which the cryptography cannot easily be swapped out.

Zero Trust Architecture, introduced in Section III, is the most consistent response to this in the 6G and IoT literature. Rather than a one-time session establishment, it requires continuous, context-aware re-verification of every device and connection [10], [11]. For vehicles, the long service life is the central problem. A 2025 survey of PQC migration across in-vehicle networks, OEM communications, and V2X finds that classical ECC-based schemes such as ECDSA still form the basis of most automotive PKI, and identifies the lack of standardized, crypto-agile designs for constrained Electronic Control Units (ECUs) as an open gap [19]. For IoT specifically, the combination of AI-driven automation for anomaly detection and key rotation with blockchain-based decentralized certificate management described in Section IV is being proposed as a way to manage device populations too large for manual key administration, though, as noted earlier, these approaches still lack mature formal security analysis [18].

Table IV summarizes the trade-offs. No single method dominates. Classical PKI is cheap and well understood but not quantum-safe and struggles to scale to IoT device counts. Hybrid PQC directly addresses HNDL risk at the cost of larger keys and more computation. Blockchain-based decentralized PKI improves resilience against a compromised CA but adds consensus latency. Zero Trust improves the scalability of trust decisions but does not, by itself, replace the underlying key exchange.

Two points recur across this part of the literature. First, crypto-agility, meaning the ability to swap cryptographic primitives without a full protocol or hardware redesign, is treated less as a convenience and more as a requirement, since constrained devices such as vehicular ECUs and industrial controllers cannot be patched as quickly as browsers or servers [19]. Second, moving away from a single centralized CA toward federated identity or blockchain-anchored certificates reduces some insider-attack and cross-domain trust problems, but it introduces a real latency cost from consensus that has not been eliminated in the systems surveyed here [18].

Looking ahead, three directions stand out. Quantum-safe migration for vehicular and industrial systems needs lightweight PQC variants tuned for constrained hardware, not simply a drop-in replacement of desktop-class algorithms [19]. Federated and decentralized identity models need clearer answers on how consensus latency is kept acceptable at IoT scale before they can be considered mature alternatives to centralized PKI [18]. Finally, Zero Trust deployments need continuous-authentication mechanisms that do not themselves become the bottleneck in latency-sensitive settings such as V2X safety messaging [11].

VI. CONCLUSION

Key distribution has come a long way from courier-delivered symmetric keys, but the underlying problem, getting two or more parties to share a secret they can trust, has not fundamentally changed. What has changed is the environment: Internet-scale deployment, device populations in the billions, and a credible, if not yet realized, quantum threat to the discrete-logarithm and factoring assumptions that Diffie-Hellman and RSA rely on.

Across the literature surveyed here, three themes recur. First, authentication and confidentiality are separable problems, and the strongest results, the Bellare-Canetti-Krawczyk authenticators and the Canetti-Krawczyk SK-security model, come from

treating them that way and composing solutions rather than reinventing them per protocol [2], [3]. Second, the current wave of post-quantum, quantum, blockchain, and AI-assisted techniques is best understood as additive rather than a replacement cycle: production TLS pairs classical ECDH with ML-KEM instead of dropping one for the other [14], and QKD deployments increasingly authenticate their classical channel with PQC instead of treating QKD as self-contained [17]. Third, and least resolved, the newer techniques, blockchain-based key management and AI-assisted key-lifecycle automation in particular, are still operating without the kind of reduction-based security proofs that classical and post-quantum key exchange now have. Closing that gap, more than any single new algorithm, looks like the most interesting open problem for the next few years of this field.

REFERENCES

- [1] W. Diffie and M. E. Hellman, "New directions in cryptography," *IEEE Trans. Inf. Theory*, vol. 22, no. 6, pp. 644–654, Nov. 1976.
- [2] M. Bellare, R. Canetti, and H. Krawczyk, "A modular approach to the design and analysis of authentication and key exchange protocols," in *Proc. 30th Annu. ACM Symp. Theory of Computing (STOC)*, Dallas, TX, USA, 1998, pp. 419–428.
- [3] R. Canetti and H. Krawczyk, "Analysis of key-exchange protocols and their use for building secure channels," in *Advances in Cryptology (EUROCRYPT 2001)*, LNCS vol. 2045. Berlin, Germany: Springer, 2001, pp. 453–474.
- [4] C. Meadows, "Analysis of the Internet Key Exchange protocol using the NRL Protocol Analyzer," in *Proc. IEEE Symp. Security and Privacy*, Oakland, CA, USA, 1999, pp. 216–231.
- [5] B. C. Neuman and T. Ts'o, "Kerberos: An authentication service for computer networks," *IEEE Commun. Mag.*, vol. 32, no. 9, pp. 33–38, Sep. 1994.
- [6] D. Cooper, S. Santesson, S. Farrell, S. Boeyen, R. Housley, and W. Polk, "Internet X.509 Public Key Infrastructure certificate and certificate revocation list (CRL) profile," IETF RFC 5280, May 2008.
- [7] B. Laurie, A. Langley, and E. Kasten, "Certificate Transparency," IETF RFC 6962, Jun. 2013.
- [8] E. Rescorla, "The Transport Layer Security (TLS) protocol version 1.3," IETF RFC 8446, Aug. 2018.
- [9] E. Barker, "Recommendation for key management: Part 1, General," NIST Special Publication 800-57 Part 1 Rev. 5, Gaithersburg, MD, USA, May 2020.
- [10] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, "Zero Trust Architecture," NIST Special Publication 800-207, Gaithersburg, MD, USA, Aug. 2020.
- [11] N. Nahar, K. Andersson, O. Schelén, and S. Saguna, "A survey on zero trust architecture: Applications and challenges of 6G networks," *IEEE Access*, vol. 12, pp. 94753–94764, 2024.
- [12] National Institute of Standards and Technology, "NIST releases first 3 finalized post-quantum encryption standards," NIST News, Aug. 13, 2024.
- [13] National Institute of Standards and Technology, "Module-lattice-based key-encapsulation mechanism standard," FIPS PUB 203, Gaithersburg, MD, USA, Aug. 2024.
- [14] D. Stebila, S. Fluhrer, and S. Gueron, "Hybrid key exchange in TLS 1.3," IETF Internet-Draft, draft-ietf-tls-hybrid-design, 2025.
- [15] P. Dehingia and N. Dutta, "Hybrid quantum key distribution framework: Integrating BB84, B92, E91, and GHZ protocols for enhanced cryptographic security," *Concurrency Comput.: Pract. Exper.*, vol. 37, no. 21–22, art. e70221, 2025.
- [16] "Field-trial quantum key distribution with qubit-based frame synchronization," arXiv preprint arXiv:2510.17659, 2025.
- [17] L.-J. Wang, K.-Y. Zhang, J.-Y. Wang, *et al.*, "Experimental authentication of quantum key distribution with post-quantum cryptography," *npj Quantum Inf.*, vol. 7, art. 67, 2021.
- [18] J. Mosakheil and K. Yang, "PKChain: Compromise-tolerant and verifiable public key management system," *IEEE Internet Things J.*, vol. 12, no. 3, pp. 3130–3144, Feb. 2025.
- [19] N. Lohmiller, S. Kaniewski, M. Menth, and T. Heer, "A survey of post-quantum cryptography migration in vehicles," *IEEE Access*, vol. 13, pp. 10160–10176, 2025.